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# chatbot_ml.py
import pandas as pd
import random
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

data = {
    'Text': [
        'Hello',
        'Hi',
        'How are you?',
        'What is your name?',
        'Bye',
        'Goodbye',
        'Thanks',
        'Thank you',
        'What can you do?',
        'Tell me a joke'
    ],
    'Intent': [
        'greeting',
        'greeting',
        'smalltalk',
        'ask_name',
        'goodbye',
        'goodbye',
        'thank_you',
        'thank_you',
        'capabilities',
        'joke'
    ]
}

df = pd.DataFrame(data)

vectorizer = TfidfVectorizer()

X = vectorizer.fit_transform(df['Text'])

y = df['Intent']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = LogisticRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy * 100:.2f}%")

responses = {
    'greeting': [
        'Hello! How can I help you today?',
        'Hi there! How can I assist you?',
        'Hello! What would you like to talk about today?'
    ],
    'smalltalk': [
        'I am a chatbot. I am always doing great!',
        'I am doing well, thanks for asking!'
    ],
    'ask_name': [
        'I am a friendly chatbot created to assist you.'
    ],
    'goodbye': [
        'Goodbye! Have a great day!',
        'See you later!',
        'Take care! Looking forward to chatting again!'
    ],
    'thank_you': [
        'You are welcome!',
        'Glad to help!'
    ],
    'capabilities': [
        'I can assist you with various tasks like answering questions or telling jokes.'
    ]
}

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],
'joke': [
    'Why don't scientists trust atoms? Because they make up everything!',
    'Why was the math book sad? Because it had too many problems.',
    'What do you call fake spaghetti? An impasta!',
    'Why don't skeletons fight each other? They don't have the guts!',
    'Why did the bicycle fall over? Because it was two-tired!'
],
'weather': [
    'It's sunny and bright outside today!',
    'Looks like it might rain later. Don't forget your umbrella!',
    'The weather is quite chilly. Stay warm!'
],
'quote': [
    '"The only limit to our realization of tomorrow is our doubts of today." - FDR',
    '"Success is not final, failure is not fatal: It is the courage to continue that counts." - Winston Churchill',
    '"Believe you can and you're halfway there." - Theodore Roosevelt'
],
'news': [
    'Here are today's top stories...',
    'In today's news, major developments have occurred in...',
    'The stock market has shown significant changes today, stay informed...'
],
'positive_feedback': [
    'Thank you! I'm happy to hear that!',
    'I'm glad you liked it!',
    'That means a lot to me!'
],
'negative_feedback': [
    'I'm sorry to hear that. I'll try to improve!',
    'Thanks for your feedback. How can I make it better?',
    'I'll do my best to get better!'
],
'fallback': [
    'I'm sorry, I didn't quite catch that. Could you try rephrasing?',
    'I'm not sure I understand. Can you say that in a different way?'
]
}

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def get_response(intent):
    return random.choice(responses[intent])

def chatbot():
    print("Chatbot: Hi! I am your friendly chatbot. Type 'exit' to end the conversation.")
    while True:
        user_input = input("You: ")

        if user_input.lower() == 'exit':
            print("Chatbot: Goodbye!")
            break

        user_input_vector = vectorizer.transform([user_input])
        predicted_intent = model.predict(user_input_vector)[0]

        bot_response = get_response(predicted_intent)
        print(f"Chatbot: {bot_response}")

if __name__ == "__main__":
    chatbot()

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*** Model Accuracy: 0.00%
Chatbot: Hi! I am your friendly chatbot. Type 'exit' to end the conversation.
You: Hello
Chatbot: Hi there! How can I assist you?
You: What is weather
Chatbot: See you later!
You: Weather
Chatbot: Goodbye! Have a great day!
You: Hello?
Chatbot: Hello! How can I help you today?
You: Good morning.
Chatbot: Take care! Looking forward to chatting again!
You: How are you

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Chatbot: I am a chatbot. I am always doing great!
You: What's the chatbot's name?
Chatbot: Goodbye! Have a great day!
You: What can you do
Chatbot: Glad to help!
You: Can you make me laugh?
Chatbot: Glad to help!
You: Can you make me laugh?
Chatbot: Glad to help!
You: Tell me a joke.
Chatbot: Why don't scientists trust atoms? Because they make up everything!
You: What's the weather like?
Chatbot: Goodbye! Have a great day!
You: Tell me the latest news.
Chatbot: What do you call fake spaghetti? An impasta!
You: